

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Deep Learning

COURSE CODE: MLA04

COURSE OUTCOME COVERED

CO1: Learning Algorithms, Maximum Likelihood Estimation (MLE), Building Machine Learning Algorithms, Neural Networks, Artificial Neurons, Perceptron, Multilayer Perceptron (MLP), Forward Propagation, Backpropagation Algorithm, Gradient Descent, Stochastic Gradient Descent (SGD), Mini-Batch Gradient Descent, Curse of Dimensionality. (BL3)

QUESTIONS: 1

Duration: 1 Hour

Total Marks:50

Annexure A

Error Analysis Task

1. A neural network predicts student scores with an MSE of 0.85 during training and 5.20 during testing. Analyse the possible reasons for this difference and suggest suitable corrective measures.

Code of Conduct:

I GUZHALINITHA.K - 192425232 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5)	Level 3 – Proficient (4)	Level 2 – Developing (2-3)	Level 1 – Beginning (0-1)
Error Identification	30%	Correctly identifies all errors and their causes	Identifies most errors with minor omissions	Identifies some errors but misses key issues	Unable to identify errors
Error Analysis & Reasoning	30%	Provides clear and logical explanation of why errors occurred	Reasoning mostly corrects with minor gaps	Limited reasoning and partial explanation	No meaningful analysis
Correction Strategy	25%	Suggests appropriate corrections with justification	Suggests correct corrections with minor gaps	Partially correct corrections	Incorrect or no corrections
Interpretation of Results	15%	Clearly explains impact of errors on model performance/results	Basic interpretation with minor omissions	Limited interpretation	No interpretation provided

Answer

1. Error Identification (30%)

The training Mean Squared Error (MSE) is **0.85**, while the testing MSE is **5.20**. This large difference indicates that the neural network performs well on the training data but poorly on unseen testing data. This problem is called **overfitting**.

Possible causes:

- The model has learned the training data too closely.
 - The training dataset is too small or not diverse.
 - The neural network has too many hidden layers or neurons.
 - Lack of regularization techniques.
 - Noise or outliers in the training data.
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2. Error Analysis and Reasoning (30%)

During training, the model minimizes the error and achieves a low MSE of **0.85**. However, when new testing data is given, the MSE increases to **5.20** because the model cannot generalize well. This happens because:

- The model memorizes the training examples instead of learning general patterns.
- Complex models capture unnecessary details from the training data.
- The testing data may have different characteristics from the training data.

Therefore, the high testing error shows **poor generalization performance**.

3. Correction Strategy (25%)

The following methods can reduce the testing error:

- **Increase the training dataset** to improve learning.
 - **Use regularization (L1/L2)** to reduce overfitting.
 - **Apply Dropout** to prevent the network from relying on specific neurons.
 - **Use Early Stopping** to stop training before overfitting occurs.
 - **Reduce the model complexity** by using fewer hidden layers or neurons.
 - **Normalize the input data** and remove noisy or incorrect data.
 - **Use Cross-Validation** to evaluate the model more effectively.
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4. Interpretation of Results (15%)

- **Training MSE = 0.85** indicates very good performance on the training dataset.
 - **Testing MSE = 5.20** indicates poor performance on new data.
 - The large difference between training and testing errors confirms **overfitting**.
 - After applying corrective measures, the testing MSE should decrease, and the gap between training and testing errors should become smaller.
 - This results in a more accurate and reliable neural network for predicting student scores.
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Conclusion

The neural network is affected by **overfitting**, which causes a low training error but a high testing error. Techniques such as **regularization, dropout, early stopping, increasing training data, reducing model complexity, and cross-validation** can improve the model's generalization ability and reduce the testing MSE. This leads to better prediction accuracy on unseen student data.